Project Summary: Spectral POD Analysis of Flow Data

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1. Motivation and Theoretical Basis

Standard Proper Orthogonal Decomposition (POD) identifies the most energetic spatial structures but ignores temporal evolution. **Spectral POD (SPOD)** addresses this by retaining both spatial and temporal information. By applying a temporal Fourier transform to the space-time correlation tensor, SPOD yields the **Cross-Spectral Density (CSD)** tensor $S(\boldsymbol{x}, \boldsymbol{x}', f)$. This allows for the extraction of modes that are coherent in both space and time, effectively isolating physically meaningful phenomena like shedding vortices or traveling waves at specific frequencies.

2. Statistical Estimation and Welch's Method

Estimating the CSD requires an ensemble of realizations. Since simulations often yield a single continuous time series, **Welch's Method** is employed. The time series is divided into N_{blk} overlapping blocks, each independently Fourier-transformed. This provides a compromise between frequency resolution and statistical noise reduction. For a selected frequency f_j, Fourier coefficients are gathered into the matrix:

$$X(f_j) = [\hat{\boldsymbol{q}}^{(1)}(f_j), \hat{\boldsymbol{q}}^{(2)}(f_j), \ldots, \hat{\boldsymbol{q}}^{(N_{\text{blk}})}(f_j)] \in \mathbb{C}^{3N \times N_{\text{blk}}}$$

3. The Role of the Weight Matrix W

The weight matrix W defines the inner product for the decomposition. In this incompressible flow study, W acts as a **discrete spatial quadrature operator**. It accounts for the volume of influence of each grid point, particularly for the non-uniform vertical (z) direction. This ensures that the extracted modes represent true physical contributions to turbulent kinetic energy rather than grid density artifacts.

4. Complete Derivation of the Method of Snapshots with Weight Matrix

To apply the Method of Snapshots for a non-uniform grid, we must track how the grid weight matrix W influences both the temporal eigenvalue problem and the final spatial mode reconstruction.

The theoretical weighted SPOD eigenvalue problem at a specific frequency f_j is given by:

$$\hat{C} W \hat{\Phi} = \hat{\Phi} \hat{\Lambda} \tag{1}$$

Substituting the definition of the CSD matrix ($\hat{C} = \frac{1}{N_{\text{blk}}} X X^*$), the system becomes:

$$\left(\frac{1}{N_{\text{blk}}} X X^* \right) W \hat{\Phi} = \hat{\Phi} \hat{\Lambda} \tag{2}$$

Because XX^*W is an impractically large $3N \times 3N$ asymmetric matrix, this system cannot be solved efficiently by direct decomposition. Multiply both the RHS and LHS by $X^*(f_j)W$:

$$X^* W \left(\frac{1}{N_{\text{blk}}} X X^* W \hat{\Phi} \right) = X^* W \hat{\Phi} \hat{\Lambda} \tag{3}$$

Using matrix multiplication associativity:

$$\frac{1}{N_{\text{blk}}} \left(X^* W X \right) \left(X^* W \hat{\Phi} \right) = \left(X^* W \hat{\Phi} \right) \hat{\Lambda} \tag{4}$$

Let $R = \frac{1}{N_{\text{blk}}} X^* W X$ and a new set of eigenvectors $\Psi = X^* W \hat{\Phi}$.

$$R \Psi = \Psi \hat{\Lambda} \tag{5}$$

This reduced system is efficient because R is a small $N_{\text{blk}} \times N_{\text{blk}}$ matrix. Also, the physical influence of the grid weights (W) is embedded within R. Solving this system yields the modal energies ($\hat{\Lambda}$) and the orthonormal eigenvectors ($\Psi^* \Psi = I$).

Reconstructing ϕ

Because X^*W is a rectangular matrix, we cannot use basic algebra to invert the definition $\Psi = X^* W \hat{\Phi}$ to find the spatial modes. Instead, following Sirovich's assumption, we construct the spatial modes as a linear combination of the original data snapshots (X), using the eigenvectors (Ψ). Our unnormalized raw modes are:

$$\hat{\Phi}_{\text{raw}} = X \Psi \tag{6}$$

Orthonormality Scaling

It is essential that our modes satisfy the weighted discrete orthonormality condition: $\hat{\Phi}^* W \hat{\Phi} = I$. Testing modes against this condition:

$$\hat{\Phi}^*_{\text{raw}} W \hat{\Phi}_{\text{raw}} = (X\Psi)^* W (X\Psi) = \Psi^* (X^* W X \Psi) \tag{7}$$

From the equation (4), we know that $X^* W X \Psi = N_{\text{blk}} \Psi \hat{\Lambda}$. Substituting this yields:

$$\Psi^* (N_{\text{blk}}) \Psi \hat{\Lambda} = N_{\text{blk}} \Psi^* \Psi \hat{\Lambda} = N_{\text{blk}} \hat{\Lambda} \tag{8}$$

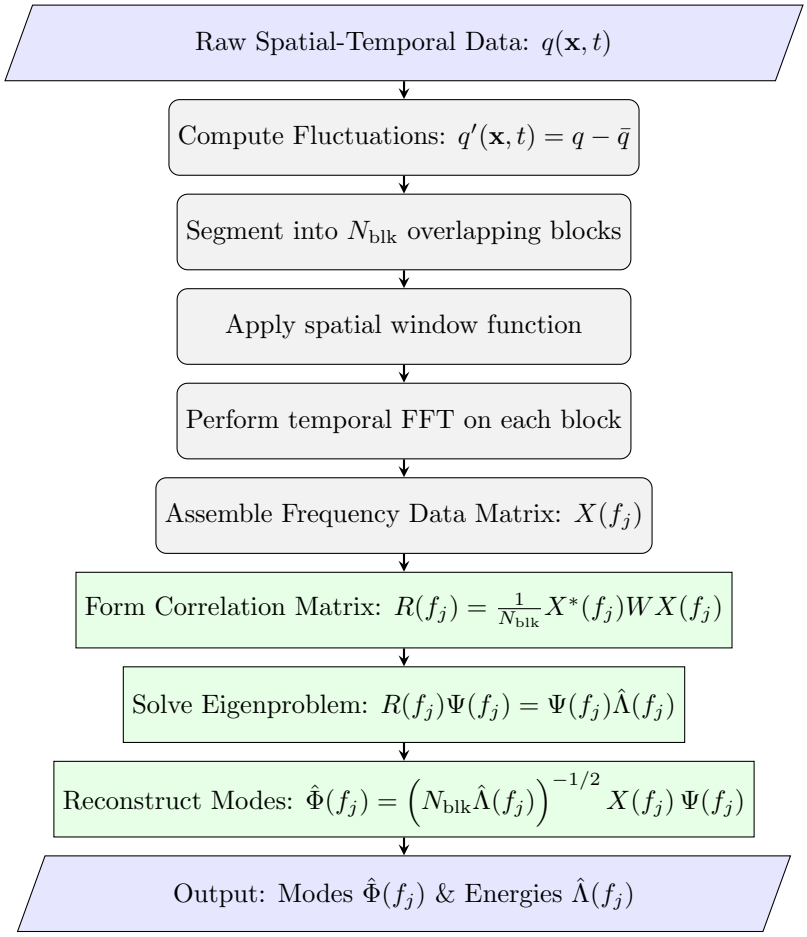
$$\hat{\Phi}^*_{\text{raw}} W \hat{\Phi}_{\text{raw}} = N_{\text{blk}} \hat{\Lambda} \tag{9}$$

This means length squared of modes is scaled by factor of $N_{\text{blk}} \hat{\Lambda}$. Because the orthonormality condition evaluates the mode quadratically, the final, properly scaled spatial SPOD modes are:

$$\hat{\Phi} = \frac{1}{\sqrt{N_{\text{blk}} \hat{\Lambda}}} X \Psi \tag{10}$$

From equation (5) we get $\hat{\Lambda}$ and from equation (10) we get our modes $\hat{\Phi}$.

Summary of the SPOD Procedure



5. Computational Implementation of SPOD

This section outlines the computational procedure used to execute the Spectral Proper Orthogonal Decomposition (SPOD) workflow. The procedure bridges the theoretical formulation with a memory-efficient discrete implementation, divided into data ingestion, preprocessing, spectral estimation, and mode reconstruction.

5.1 Data Ingestion and Matrix Setup

The raw Direct Numerical Simulation (DNS) data, originally generated and written by Fortran using unformatted stream access (`ACCESS='STREAM'`), exists as a continuous stream of binary bytes without record markers. Because the raw input is a sequential 1D stream, and the SPOD algorithm requires rapid, non-sequential access to overlapping time segments, the data must be structurally rearranged. The algorithm systematically reads the raw stream block by block, identifying exactly $N_x \times N_y \times N_z$ scalar values for each of the three spatial degrees

of freedom (velocity components U, V, W) per timestep. These sequential blocks are concatenated to build the structured 2D time-domain snapshot matrix, Q, where rows represent the flattened spatial coordinates and columns represent discrete time snapshots. To manage the massive data footprint and allow the SPOD code to instantly access these time blocks, the rearranged matrix Q is written directly to the hard drive as a memory-mapped array (done using the `build_Q_memmap` function), circumventing system memory limitations during processing.

5.2 Spatial Weight Matrix Computation

In the discrete formulation of SPOD, the weight matrix W defines the inner product under which the spatial modes are optimal and orthogonal, appearing in the eigenvalue problem:

$$\hat{C}W\hat{\Phi} = \hat{\Phi}\hat{\Lambda}. \tag{11}$$

Physically, W acts as a numerical quadrature rule that accounts for the varying spatial measure of each grid point. This is critical in direct numerical simulations of boundary layers or interfacial flows, where non-uniform grid spacing is required to resolve high-gradient regions in the vertical direction.

Without a spatial correction, clustered nodes in fine-grid regions would disproportionately dominate the kinetic energy calculation simply due to their high numerical density. To ensure the decomposition reflects true physical energy, each grid point must be weighted by its local control volume.

For the current computational domain, the horizontal grid spacing is uniform ($\Delta x = L_x/N_x$ and $\Delta y = L_y/N_y$), while the vertical coordinates (z) are strictly increasing but non-uniform. The discrete vertical weight for a node at index k is computed using a 1D trapezoidal rule:

$$\Delta z_k = \frac{1}{2}(z_{k+1} - z_{k-1}), \tag{12}$$

with standard one-sided differences applied at the domain boundaries. The local spatial weight for any node is the three-dimensional control volume, $V_{\text{node}} = \Delta x \Delta y \Delta z_k$.

Algorithmic consistency is verified by ensuring the sum of all nodal weights exactly reproduces the true physical volume of the domain:

$$\sum_k \sum_j \sum_i \Delta x \Delta y \Delta z_k = L_x L_y (z_{\max} - z_{\min}). \tag{13}$$

Finally, because the state vector consists of three dimensionally identical velocity components ($q = [u, v, w]^T$), no specialized variable-scaling is required. The full weight matrix W is constructed as a 1D array containing the diagonal elements by tiling the block of localized grid volumes across the three spatial degrees of freedom. This ensures every velocity component is scaled identically according to the physical geometry of the mesh.

5.3 Block Segmentation and Fluctuation Extraction

To estimate the robust Cross-Spectral Density (CSD), the full snapshot matrix Q is partitioned into an ensemble of shorter temporal blocks. If these blocks were extracted sequentially without overlap (a technique often referred to as the Bartlett method), the abrupt truncation of the time-series signal at the block boundaries would introduce artificial high-frequency noise into the Fourier transform, a phenomenon known as spectral leakage. To mitigate this, a temporal windowing function—specifically, a Hann window—is applied to each block. The Hann window acts as a mathematical envelope that smoothly tapers the velocity signal to zero at both the beginning and the end of the segment, enforcing strict periodicity and eliminating leakage.

However, the Hann window introduces a secondary complication: by forcing the boundaries to zero, it effectively discards the physical data located near the edges of every block. To compensate for this systematic data loss, the algorithm employs Welch's method by allowing the segments to overlap by 50%. This overlap ensures that physical turbulence attenuated at the edge of one block sits squarely at the unattenuated peak of the adjacent block, preserving 100% of the simulation's energy.

Furthermore, by dividing the total time series into these overlapping segments, the algorithm significantly increases the number of available block realizations. This overlapping ensemble averaging is mathematically essential; it increases the statistical degrees of freedom and reduces the variance of the final spectral estimate. In short, because true turbulent structures are physically coherent across time, breaking the full dataset into many overlapping blocks and averaging their spectra together allows the random, chaotic noise to cancel itself out, while keeping the distinct energy peaks of the underlying physical phenomena intact.

The matrix is loaded into memory piece by piece by selectively extracting block-specific columns directly from the memmap. Once a block is loaded, the temporal mean of that specific block is calculated and subtracted from

every row. This critical step applies a standard Reynolds decomposition, converting the raw velocity data into purely turbulent fluctuation data.

$$q'(\mathbf{x}, t) = q(\mathbf{x}, t) - \bar{q}_{\text{block}}(\mathbf{x}), \tag{14}$$

It should be noted that performing the mean subtraction at this stage—rather than having the Fortran solver export pre-computed fluctuations—is a deliberate architectural choice. Configuring the DNS to export the full, instantaneous velocity field (q) preserves the complete physical state of the flow. This maximizes the dataset's flexibility, allowing the same binary output to be used for different types of analyses in the future without requiring the expensive simulation to be re-run. Consequently, the SPOD pipeline is responsible for isolating the fluctuations.

From a fluid dynamics perspective, SPOD is designed to extract dynamic, coherent turbulent structures interacting at specific, non-zero frequencies. If the stationary mean flow ($\bar{q}$) were retained, its massive kinetic energy would completely dominate the decomposition, manifesting as an overwhelming zero-frequency ($f = 0$) mode. By subtracting the block mean, the algorithm isolates turbulent fluctuations relative to the instantaneous baseline velocity profile. This preprocessing directs the decomposition toward transient eddy structures at non-zero frequencies. The residual energy visible at ($f = 0$) reflects block-to-block variations in the mean flow that persist despite this preprocessing, a feature expected when the baseline state is not strictly stationary..

5.4 Spectral Transformation and Frequency Matrix Assembly

Following the temporal preprocessing, the algorithm transforms the data from the time domain into the frequency domain. For every overlapping block, a discrete Real Fast Fourier Transform (FFT) is applied along the temporal dimension. To preserve physical energy, the resulting Fourier coefficients are scaled by the factor:

$$\kappa = \sqrt{\frac{\Delta t}{\sum_{n=1}^{N_{\text{blk}}} w_n^2}}, \tag{15}$$

where w_n is the amplitude of the Hann window at sample n and Δt is the snapshot time spacing. This scaling compensates for the variance attenuation introduced by the windowing and ensures that the final eigenvalues $\hat{\Lambda}(f)$ represent power spectral density with correct physical dimensions.

In classical SPOD theory, applying the FFT across all blocks yields a massive three-dimensional tensor (Space $\times$ Blocks $\times$ Frequencies). However, holding this full tensor in system memory is computationally prohibitive for high-fidelity DNS datasets. To circumvent this, the algorithm is structured to evaluate the decomposition one frequency at a time.

For a targeted discrete frequency, f_j, the algorithm extracts the corresponding complex Fourier coefficients from every block and concatenates them into a localized 2D frequency-domain snapshot matrix, $X(f_j) \in \mathbb{C}^{N_{\text{space}} \times N_{\text{blk}}}$. In this matrix, rows represent the flattened spatial degrees of freedom, and each column represents the realization of that specific frequency from a single temporal block. This isolated matrix $X(f_j)$ contains all the necessary spatial and statistical information required to compute the reduced correlation matrix (introduced in the next subsection) for the chosen frequency, dramatically reducing the memory footprint of the subsequent eigenvalue problem.

5.5 Assembly of the Reduced Matrix with Chunked Weight Integration

The computational pipeline is explicitly designed to solve the reduced eigenvalue problem derived in Section 4. Specifically, this step constructs the reduced correlation matrix R and solves for the modal energies ($\hat{\Lambda}$) and temporal eigenvectors (Ψ) as defined in Equation (5).

However, while the theoretical derivation for R (detailed in Equation 4) assumes a monolithic matrix multiplication to evaluate the weighted inner product X^*WX, computing this simultaneously for the entire spatial domain significantly exceeds standard system memory limits.

To circumvent this, the algorithm exploits the distributive property of matrix multiplication. The massive spatial domain is divided into manageable segments (e.g., chunks of 20,000 grid points). For each spatial chunk, the algorithm extracts the corresponding rows of the frequency matrix $X(f_j)$, multiplies them by their localized spatial weights from W, and computes the partial inner product. These partial matrices are cumulatively summed across all spatial chunks and finally ensemble-averaged by the block count (N_{blk}). This chunking strategy mathematically yields the exact R matrix defined in Section 4 without ever loading the full $X(f_j)$ matrix into memory simultaneously.

Finally, while R is theoretically guaranteed to be perfectly Hermitian, calculating billions of chunked floating-point multiplications introduces microscopic numerical roundoff errors. To ensure strict numerical stability for the eigenvalue solver, the algorithm explicitly enforces Hermitian symmetry on the final matrix by averaging it with its conjugate transpose ($R = \frac{1}{2}(R + R^*)$) before proceeding to the decomposition step.

5.6 Reduced Eigenvalue Decomposition

Once the reduced Cross-Spectral Density matrix $R(f_j)$ is constructed and stabilized, the algorithm solves the Hermitian eigenvalue problem defined in Equation (5).

Using an optimized linear algebra solver, the pipeline extracts the discrete modal energies ($\hat{\Lambda}$) and their corresponding orthonormal temporal eigenvectors (Ψ). Because the primary objective of SPOD is to isolate the most dominant coherent flow structures, the algorithm immediately sorts the resulting eigenvalues (and their corresponding eigenvectors) in strictly descending order. Consequently, the first mode ($\hat{\Phi}_1$) is mathematically guaranteed to represent the single most energetic turbulent phenomenon at the targeted frequency f_j.

Finally, a numerical tolerance filter is applied to the eigenvalues. While true physical kinetic energy must be strictly non-negative, the billions of floating-point operations required to build the R matrix can introduce microscopic, non-physical negative eigenvalues (e.g., on the order of machine epsilon, 10^{-16}). The algorithm identifies these numerical artifacts and explicitly forces them to zero, ensuring physical validity before the spatial reconstruction phase.

5.7 Spatial Mode Reconstruction and Verification

With the modal energies ($\hat{\Lambda}$) and temporal eigenvectors (Ψ) successfully extracted, the pipeline proceeds to the final step: reconstructing the physical 3D spatial modes ($\hat{\Phi}$). Because the spatial mapping matrix X^* is rectangular, the eigenvectors cannot be algebraically inverted. Instead, the algorithm constructs the spatial modes as a linear combination of the original frequency-domain snapshots, projected onto the temporal eigenvectors. To guarantee that these final spatial modes satisfy the weighted discrete orthonormality condition ($\hat{\Phi}^* W \hat{\Phi} = I$), the projection is scaled according to the derivation in Equation (10). Just as with the correlation matrix assembly, computing $\hat{\Phi} = \frac{1}{\sqrt{N_{\text{blk}}\hat{\Lambda}}} X\Psi$ requires a matrix multiplication that exceeds available system memory. Therefore, the algorithm once again utilizes a memory-mapped, chunked matrix multiplication. The pipeline reads the frequency matrix $X(f_j)$ in spatial segments, projects it onto the scaled eigenvectors, and streams the resulting spatial modes directly to the hard drive. As a final rigorous validation step, the algorithm computationally verifies the orthonormality of the reconstructed modes. The code performs a localized, chunked computation of $\hat{\Phi}^* W \hat{\Phi}$ and compares the result to an exact identity matrix. Verifying that the maximum residual error approaches machine zero (e.g., $\sim 10^{-15}$) guarantees that the extracted modes are physically independent, orthogonal, and correctly scaled with respect to the non-uniform grid volume.

5.8 Physical Interpretation: Eigenvalues as Power Spectral Density

The eigenvalues $\hat{\Lambda}_k(f_j)$ computed at each frequency f_j represent the contribution of the k-th spatial mode to the kinetic energy spectrum at that frequency. Because the Fourier coefficients were scaled by the window-compensation factor κ (defined in Section 5.4), the eigenvalues inherit the physical dimension of power spectral density: energy per unit frequency.

The sum $\sum_{k=1}^{N_{\text{blk}}} \hat{\Lambda}_k(f_j)$ yields the total kinetic energy density at frequency f_j. Integrating this quantity across all frequencies recovers the total turbulent kinetic energy in the flow field:

$$\text{TKE} = \sum_{f_j} \sum_{k=1}^{N_{\text{blk}}} \hat{\Lambda}_k(f_j)\, \Delta f, \tag{16}$$

where Δf is the frequency resolution. In this way, the SPOD spectrum directly quantifies how kinetic energy is distributed across both spatial scales (through the mode index k) and temporal scales (through the frequency axis). The dominant eigenvalue at the peak frequency reveals not only where in the spectrum the strongest coherent structures live, but also their energetic importance relative to the total turbulent cascade.

5.9 Full Spectrum Execution and Energy Aggregation

With the single-frequency pipeline fully optimized and validated, the algorithm iterates over the complete set of discrete physical frequencies generated by the Real FFT. The physical frequency axis is explicitly defined by the sampling rate of the DNS snapshots (Δt). To ensure continuous execution without exceeding system memory limits or encountering operating system file-lock conflicts, the pipeline explicitly purges memory-mapped file handles and triggers Python's garbage collection at the start of each frequency iteration. Furthermore, the rigorous spatial orthonormality verification ($\hat{\Phi}^* W \hat{\Phi} = I$) is restricted to the initial frequency step to minimize computational overhead, as the mathematical structure of the projection is invariant across frequencies. As the pipeline iterates through the spectrum, the algorithm extracts the localized 3D spatial modes ($\hat{\Phi}$) directly to the hard drive, while simultaneously aggregating the modal energies ($\hat{\Lambda}$) into a global data structure. This global aggregation yields the complete SPOD Energy Spectrum, which serves as the primary diagnostic tool for

identifying the dominant, coherent turbulent structures and their respective shedding frequencies within the flow field.

5.10 Spectral Energy Distribution and Mode Dominance

Following the SPOD computation, the resulting modal energy distribution was analysed across the discrete frequency bins using three diagnostic ratios: the leading-mode energy $\hat{\Lambda}_1$, the mode-1 coherence ratio $\hat{\Lambda}_1/\sum_k \hat{\Lambda}_k$, and the rank separation $\hat{\Lambda}_1/\hat{\Lambda}_2$. The total summed modal energy across the spectrum is $\sum_f \sum_k \hat{\Lambda}_k(f) = 2.02 \times 10^1$. As shown in Table 1, the spectrum exhibits a sharp global peak at $f = 0.02$.

Table 1: Top five frequencies ranked by leading mode energy $\hat{\Lambda}_1$. The $f = 0$ entry reflects residual block-to-block variation in the mean velocity field rather than a true zero-frequency mode, since each block undergoes independent mean subtraction prior to the FFT.

Rank	f	$\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1$	$\hat{\Lambda}_1/\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1/\hat{\Lambda}_2$	**Modes** $\geq 90\%$
1	0.0200	3.474	1.178	0.339	1.97	10
2	0.0400	3.079	0.466	0.151	1.14	14
3	0.0000	1.655	0.359	0.217	1.21	8
4	0.0600	2.256	0.261	0.116	1.19	16
5	0.0800	1.719	0.254	0.148	1.53	16

At the peak frequency $f = 0.02$, the leading mode captures 33.9% of the local spectral energy, and the rank separation $\hat{\Lambda}_1/\hat{\Lambda}_2 = 1.97$ is just below the ≥ 2.0 threshold conventionally used to define rank-1 dominance. The strongest separation in the entire spectrum occurs at $f = 0.12$, with $\hat{\Lambda}_1/\hat{\Lambda}_2 = 2.09$, suggesting that this frequency hosts a particularly distinct coherent structure despite its lower absolute energy. These two frequencies, together with $f = 0.06$ (the broader coherent regime) and $f = 0.40$ (a turbulent reference), define the four target frequencies analysed in detail in subsequent sections.

The mode-1 coherence ratio identifies a clear coherent–incoherent transition. With $N_{\text{blk}} = 19$ blocks, the incoherent baseline is $1/N_{\text{blk}} = 0.053$ and the conventional coherence threshold is $2/N_{\text{blk}} = 0.105$. The ratio $\hat{\Lambda}_1/\sum_k \hat{\Lambda}_k$ exceeds this threshold only for $f \leq 0.12$, with the first crossing below it occurring at $f = 0.14$. Beyond this frequency, the energy is distributed nearly uniformly across all blocks, and the number of modes required to capture 90% of the variance saturates at 16–17 — close to the maximum of $N_{\text{blk}} = 19$. This is the spectral signature of broadband, spatially uncorrelated turbulent fluctuations and defines the upper edge of the coherent regime at $f \approx 0.14$.

5.11 SPOD Eigenvalue Spectrum

Figure 1 presents the complete SPOD eigenvalue spectrum across all 26 resolved frequencies, displayed on both linear-frequency/log-energy and log–log axes.

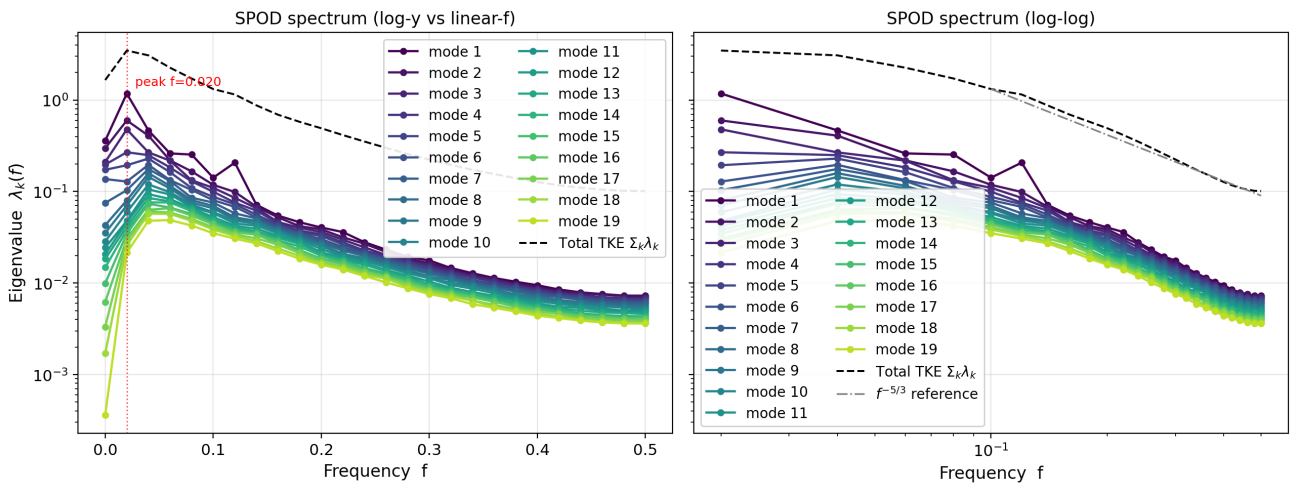


Figure 1: SPOD eigenvalue spectrum across all resolved frequencies. Each curve corresponds to a single mode rank (modes 1–19, darkest to lightest); the dashed black line is the total energy per frequency $\sum_k \hat{\Lambda}_k$. Left: log-energy versus linear frequency, with the global peak at $f = 0.02$ (red dotted line). Right: log–log axes, with a dash-dotted $f^{-5/3}$ reference line.

The spectrum reveals three distinct frequency regimes. At low frequencies ($f \leq 0.04$), the leading mode is well separated from the remaining modes, with the global energy peak at $f = 0.02$ where $\hat{\Lambda}_1 = 1.178$ and the coherence ratio reaches $\hat{\Lambda}_1 / \sum_k \hat{\Lambda}_k = 0.339$. This separation identifies a coherent structure concentrated at a single frequency. An intermediate coherent regime extends up to $f \approx 0.12$, where the leading mode retains a meaningful energy advantage over the sub-leading modes. Beyond $f \approx 0.14$, the individual mode curves converge and the spectrum transitions into an incoherent turbulent regime in which energy is distributed nearly uniformly across all 19 mode ranks — the spectral signature of broadband, spatially uncorrelated fluctuations.

A localised secondary elevation is clearly visible near $f = 0.12$ in both mode 1 and mode 2, breaking the otherwise monotonic decay of the leading-mode curve. This bump corresponds to the highest rank-separation $\hat{\Lambda}_1/\hat{\Lambda}_2 = 2.09$ in the spectrum (Section **??**) and motivates the inclusion of $f = 0.12$ as a target frequency in Section 5.12.

On the log–log axes, the total energy curve $\sum_k \hat{\Lambda}_k$ tracks the $f^{-5/3}$ reference closely from $f \approx 0.05$ up to the Nyquist frequency $f = 0.5$. This is broadly consistent with a classical Kolmogorov inertial subrange and indicates that the small-scale fluctuations remain well-resolved across the full sampled bandwidth.

5.12 Vertical Structure of the Three Leading Modes Across Frequencies

The vertical organisation of the three most energetic SPOD modes was examined as a function of frequency. Figures 2, 3, and 4 present the phase-aligned real parts of $\hat{\Phi}_1$, $\hat{\Phi}_2$, and $\hat{\Phi}_3$ at four target frequencies ($f = 0.02$, 0.06, 0.12, and 0.40), sliced at mid-span ($y = 4.737$). The four frequencies sample the three spectral regimes identified in Section 6.2: the primary coherent peak ($f = 0.02$), the intermediate coherent range ($f = 0.06$ and $f = 0.12$), and the incoherent turbulent reference ($f = 0.40$).

Canonical phase alignment

SPOD modes are complex-valued, encoding both the spatial amplitude and the convective phase of a propagating structure. Because the temporal origin of the snapshot ensemble is arbitrary, a direct extraction of the real part of $\hat{\Phi}$ would yield an arbitrary instantaneous slice of the wave cycle and would obscure direct comparison between modes at different frequencies. To visualise each mode at a physically reproducible state, a global phase rotation $e^{-i\varphi^\star}$ was applied prior to plotting, where $\varphi^\star$ is the angle that maximises the W-weighted real-part energy of the rotated mode. The closed-form solution is:

$$\varphi^\star = \frac{1}{2} \arg \left(\sum_n W_n \, \hat{\Phi}_n^2 \right), \tag{17}$$

where W_n are the diagonal entries of the spatial weight matrix introduced in Section 5.2 and the sum runs over all spatial degrees of freedom. The fraction of total mode energy carried by the real part after this alignment is a useful diagnostic: a value near 100% corresponds to a standing wave, while a value near 50% indicates a purely propagating wave, since the energy of a travelling wave is partitioned equally between its real and imaginary components. Across all twelve mode–frequency combinations considered here, the real-part energy fraction lies within 0.3% of the theoretical 50% value, confirming that every plotted structure corresponds to a downstream-propagating wave.

Mode 1: dominant coherent structure

The frequency progression of mode 1 (Figure 2) reveals a systematic transformation in spatial character. At $f = 0.02$ the structure is large and smooth, with U and W exhibiting paired positive/negative regions extending across the full streamwise extent of the box. Energy is concentrated in the lower-to-mid region, with peak at $z = 0.39$ and centroid $\bar{z} = 0.49$, and the component partition $U : V : W = 68.0\% : 21.6\% : 10.4\%$ confirms strong streamwise dominance. At $f = 0.06$ the structure contracts to roughly two streamwise periods within $L_x = 3\pi$ and develops a forward-leaning inclination visible in the V and W components, consistent with mean-shear deformation of coherent structures. The mode at $f = 0.12$ exhibits qualitatively different behaviour: U weakens substantially while V and especially W become concentrated in a thin near-surface layer ($z \lesssim 0.15$) with high streamwise wavenumber. The component partition shifts to $U : V : W = 44.1\% : 20.4\% : 35.5\%$, with W approaching parity with U — the only frequency in the spectrum at which this occurs, and the same frequency that exhibits the highest rank-separation $\hat{\Lambda}_1/\hat{\Lambda}_2 = 2.09$ in Section 6.2. At $f = 0.40$, all three components display fine-scale, spatially disorganised fluctuations distributed throughout the wall-normal extent, and the partition approaches near-isotropy ($U : V : W \approx 26\% : 41\% : 33\%$), consistent with the incoherent turbulent regime.

Modes 2 and 3: independent sub-dominant structures

At $f = 0.02$, modes 1, 2, and 3 collectively account for approximately 65% of the total spectral energy, raising the question of whether the sub-dominant modes represent genuinely independent structures or are simply variants

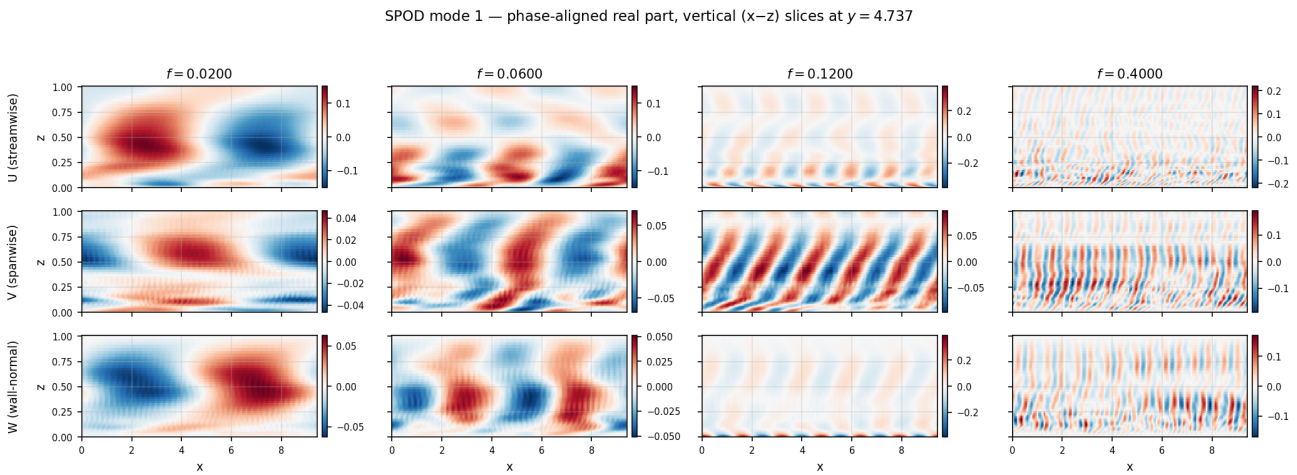


Figure 2: Phase-aligned real part of SPOD mode 1 at four frequencies, plotted as vertical (x–z) slices at mid-span ($y = 4.737$). Rows: U, V, W velocity components. Columns: increasing frequency from left to right. Each panel is normalised to its own maximum amplitude.

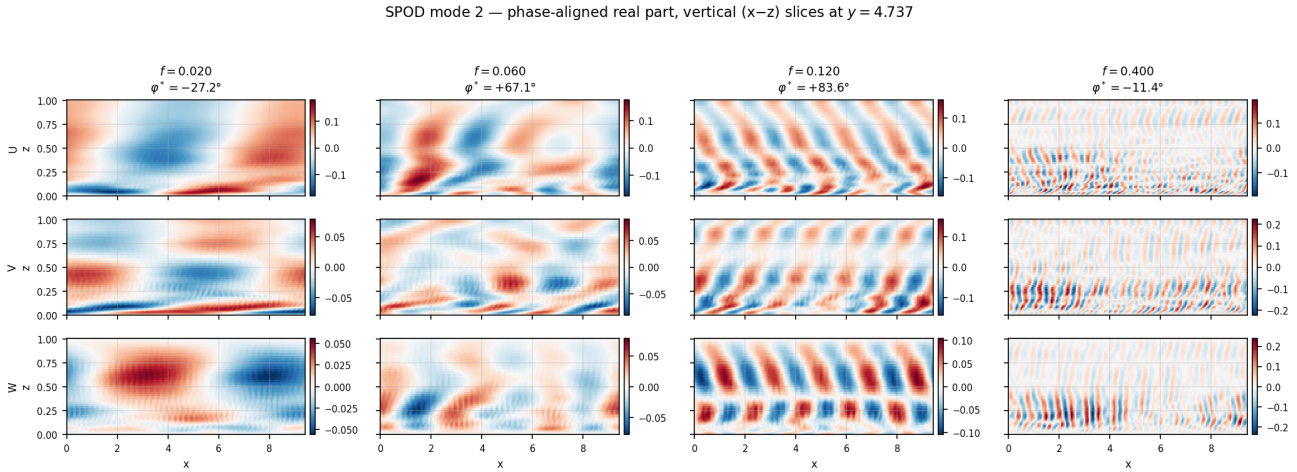


Figure 3: Phase-aligned real part of SPOD mode 2 at four frequencies, plotted as vertical (x–z) slices at mid-span. Layout identical to Figure 2.

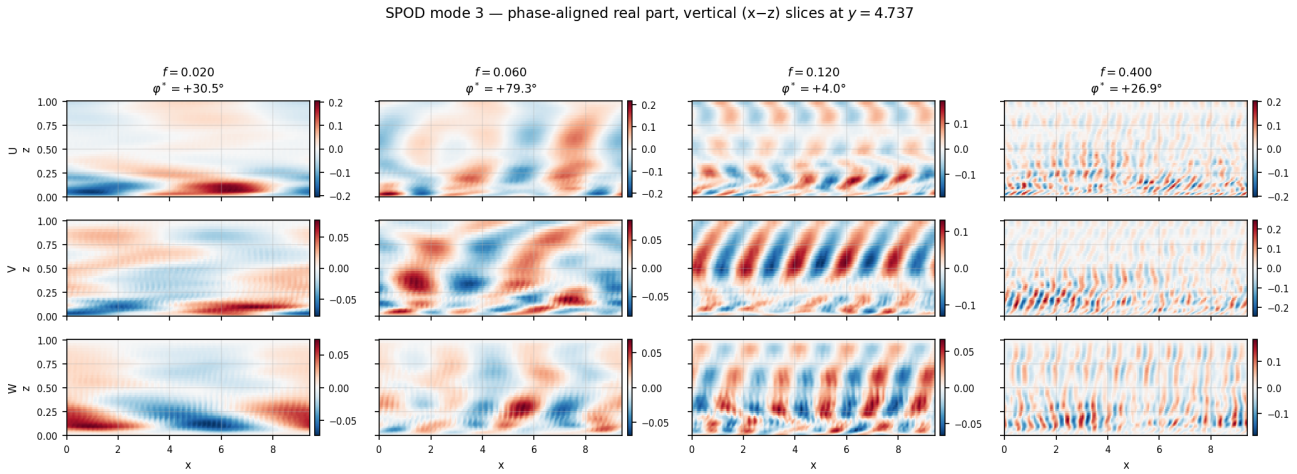


Figure 4: Phase-aligned real part of SPOD mode 3 at four frequencies, plotted as vertical (x–z) slices at mid-span. Layout identical to Figure 2.

of mode 1 (e.g. harmonics or phase-shifted copies). The two sub-dominant modes (Figures 3 and 4) differ from mode 1 in both spatial localisation and component partition. Mode 2 places its strongest U and W activity in

the upper half of the domain ($z \gtrsim 0.5$), while mode 3 is sharply concentrated in the lower quarter ($z \lesssim 0.25$). The component partitions, $U : V : W = 72.2\% : 17.7\% : 10.1\%$ for mode 2 and $U : V : W = 76.0\% : 14.6\% : 9.5\%$ for mode 3, are systematically *more* streamwise-dominated than mode 1's $68.0\% : 21.6\% : 10.4\%$. The differing wavelengths and wall-normal localisations rule out the harmonic and phase-shifted-copy hypotheses, identifying the three modes as distinct coexisting structures rather than overtones of a single underlying motion.

At higher frequencies, modes 2 and 3 follow the same general trends as mode 1: shorter streamwise wavelengths and broader wall-normal distributions in the intermediate range, and convergence to a near-identical fine-scale character at $f = 0.40$ with statistically indistinguishable component partitions ($U : V : W \approx 26$–$27\% : 40$–$41\% : 33\%$). This convergence reflects the loss of mode-specific spatial organisation in the incoherent regime.

Component-energy cascade across rank and frequency

A consistent quantitative trend emerges across the three modes. Within the coherent regime, the streamwise share U *increases* with mode rank ($68.0\% \to 72.2\% \to 76.0\%$ at $f = 0.02$), reflecting an increasingly streak-dominated character in higher-rank modes. With increasing frequency, the U share *decreases* for every individual mode and the partition approaches isotropy. By $f = 0.40$, all three modes share the same mildly V-biased near-isotropic split. The streamwise-anisotropy imprint of the wave-driven shear is therefore strongest in the leading mode at the lowest coherent frequency and is progressively erased moving toward either higher rank or higher frequency.

5.13 Wall-Normal Energy Distribution and Component-Energy Cascade

To complement the qualitative spatial visualisation of the leading SPOD modes, two quantitative diagnostics were extracted from the mode field. Panel (a) of Figure 5 presents the wall-normal energy distribution $E(z)$ of mode 1 at the four target frequencies, while panel (b) presents the streamwise, spanwise, and wall-normal component-energy shares of mode 1 across the entire resolved frequency range.

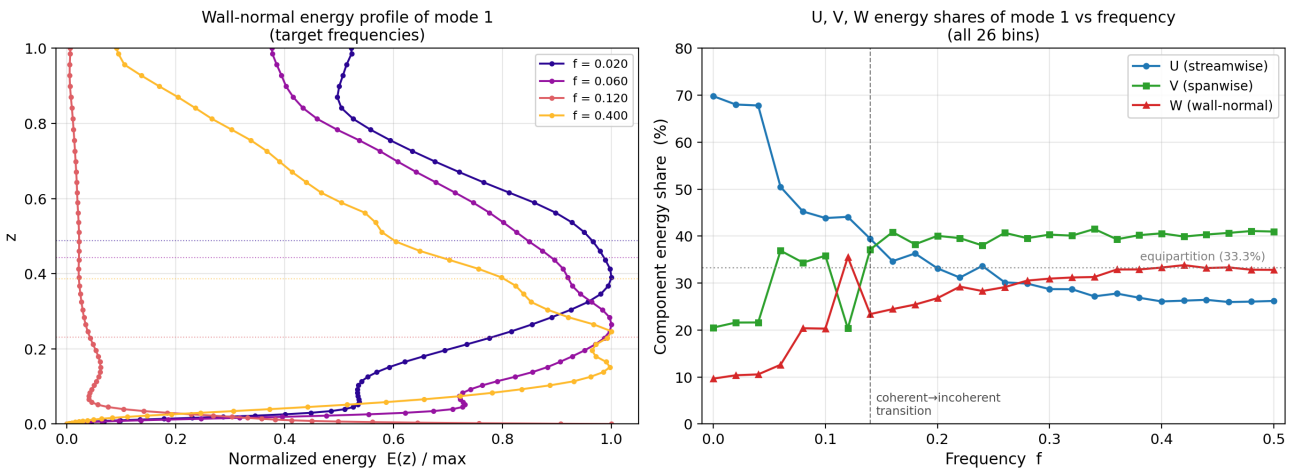


Figure 5: (a) Wall-normal energy profile of SPOD mode 1 at the four target frequencies. Profiles are normalised to their own peak and dotted horizontal lines mark the corresponding energy centroids $\bar{z}$. (b) U, V, and W component-energy shares of mode 1 plotted across all 26 resolved frequency bins. The horizontal grey dotted line marks the equipartition threshold (1/3) and the vertical dashed line marks the coherent–incoherent transition identified in Section 6.2.

Wall-normal energy profiles

The wall-normal distribution of mode 1 varies substantially between the four target frequencies, with no two profiles sharing the same shape. The relevant integral diagnostics are summarised in Table 2.

At $f = 0.02$ the profile is bimodal, with a primary maximum at $z_{\text{peak}} = 0.39$ and a near-equal secondary maximum near $z \approx 0.95$, yielding a centroid at $\bar{z} = 0.49$. The profile at $f = 0.06$ is more conventional, peaking at $z_{\text{peak}} = 0.27$ with centroid $\bar{z} = 0.44$, and the mode occupies the lower-to-middle portion of the domain. The mode at $f = 0.12$ is the only profile peaking at the wave surface itself ($z_{\text{peak}} = 0.000$) and drops by an order of magnitude within $z \lesssim 0.1$, but a small secondary elevation is nevertheless present near the upper boundary; the centroid $\bar{z} = 0.23$ confirms the strongly surface-localised nature already identified in Section 5.12. At $f = 0.40$ the profile is broader and more diffuse than the coherent profiles, with a weak maximum near $z_{\text{peak}} = 0.25$ and a secondary elevation in the upper region ($z \approx 0.9$). This diffuse distribution is consistent with the spatially incoherent nature of the small-scale fluctuations identified in Section 6.2. Each frequency therefore samples a physically distinct wall-normal organisation rather than a monotonic family of progressively detached structures.

Table 2: Wall-normal localisation of mode 1 at the four target frequencies. z_{peak} is the height of the energy maximum and $\bar{z}$ is the volume-weighted centroid; the integral $\int E(z)\,\mathrm{d}z = 1$ in all cases by construction, confirming the orthonormality of the modes.

f	z_{peak}	$\bar{z}$	$\int E(z)\,\mathrm{d}z$
0.0200	0.390	0.488	1.000
0.0600	0.265	0.443	1.000
0.1200	0.000	0.231	1.000
0.4000	0.247	0.386	1.000

Component-energy cascade across frequency

Panel (b) of Figure 5 reveals a systematic anisotropy cascade in the leading mode. The streamwise share U decreases overall from a peak of 69.8% at $f = 0$ to a minimum of 26.0% at $f = 0.46$, while the spanwise share V rises from 20.5% to a plateau of approximately 40%, and the wall-normal share W rises from 9.7% to roughly 33%. Three distinct phases of this cascade can be identified.

In the primary coherent regime ($f \leq 0.04$), the leading mode is strongly U-dominated, with the partition $U : V : W \approx 70:21:10$ characteristic of a streamwise-elongated streak structure. Across the intermediate coherent range ($0.06 \leq f \leq 0.12$), the U share drops sharply from 54% to approximately 44% while V and W both grow, with W exhibiting an isolated peak of 35.5% at $f = 0.12$ — the only frequency in the spectrum at which W exceeds V, consistent with the surface-locked, wall-normal-dominated structure identified in Section 5.12. Beyond the coherent–incoherent transition at $f \approx 0.14$, all three component shares stabilise: the U share crosses the equipartition threshold (1/3) at $f = 0.20$, indicating the loss of streamwise dominance, and from $f \approx 0.30$ onward the partition remains approximately $U : V : W \approx 27:40:33$. The mild V-bias in this asymptotic state, with V slightly exceeding W and clearly exceeding U, indicates that the small-scale fluctuations are close to but not strictly isotropic, retaining a weak anisotropy associated with the wall-bounded geometry.

The cascade summarised here provides a compact quantitative description of how energy migrates between velocity components as spatial scales decrease: the streamwise-anisotropic imprint of the wave-driven shear is strongest at the lowest coherent frequencies and is progressively erased as the cascade approaches the small-scale turbulent regime.

Appendix: Complete SPOD Eigenvalue Spectrum

Table 3: Full SPOD spectral data across all resolved frequencies ($N_f = 26$, $N_{\text{blk}} = 19$).

idx	f	$\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1$	$\hat{\Lambda}_1/\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1/\hat{\Lambda}_2$	**Modes** $\geq 90\%$
0	0.0000	1.6545e+00	3.5888e-01	0.217	1.21	8
1	0.0200	3.4743e+00	1.1780e+00	0.339	1.97	10
2	0.0400	3.0786e+00	4.6578e-01	0.151	1.14	14
3	0.0600	2.2564e+00	2.6110e-01	0.116	1.19	16
4	0.0800	1.7191e+00	2.5358e-01	0.148	1.53	16
5	0.1000	1.3179e+00	1.4172e-01	0.108	1.20	16
6	0.1200	1.1498e+00	2.0701e-01	0.180	2.09	16
7	0.1400	8.7465e-01	7.1543e-02	0.082	1.05	16
8	0.1600	6.9101e-01	5.4459e-02	0.079	1.05	17
9	0.1800	5.7765e-01	4.6100e-02	0.080	1.10	17
10	0.2000	4.9148e-01	4.0444e-02	0.082	1.05	17
11	0.2200	4.1415e-01	3.6056e-02	0.087	1.14	17
12	0.2400	3.5189e-01	2.7946e-02	0.079	1.04	17
13	0.2600	3.0017e-01	2.3142e-02	0.077	1.06	17
14	0.2800	2.5751e-01	1.9418e-02	0.075	1.05	17
15	0.3000	2.2405e-01	1.7650e-02	0.079	1.07	17
16	0.3200	1.9598e-01	1.4605e-02	0.075	1.03	17
17	0.3400	1.7270e-01	1.2801e-02	0.074	1.05	17
18	0.3600	1.5536e-01	1.1350e-02	0.073	1.03	17
19	0.3800	1.3994e-01	1.0335e-02	0.074	1.05	17
20	0.4000	1.2711e-01	9.4636e-03	0.074	1.05	17
21	0.4200	1.1709e-01	8.5011e-03	0.073	1.04	17

idx	f	$\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1$	$\hat{\Lambda}_1 / \sum \hat{\Lambda}_k$	$\hat{\Lambda}_1 / \hat{\Lambda}_2$	**Modes** $\geq 90\%$
22	0.4400	1.1012e-01	7.9072e-03	0.072	1.02	17
23	0.4600	1.0509e-01	7.5801e-03	0.072	1.01	17
24	0.4800	1.0193e-01	7.2930e-03	0.072	1.01	17
25	0.5000	1.0114e-01	7.2718e-03	0.072	1.02	17
Σ	—	2.0160e+01	—	—	—	—